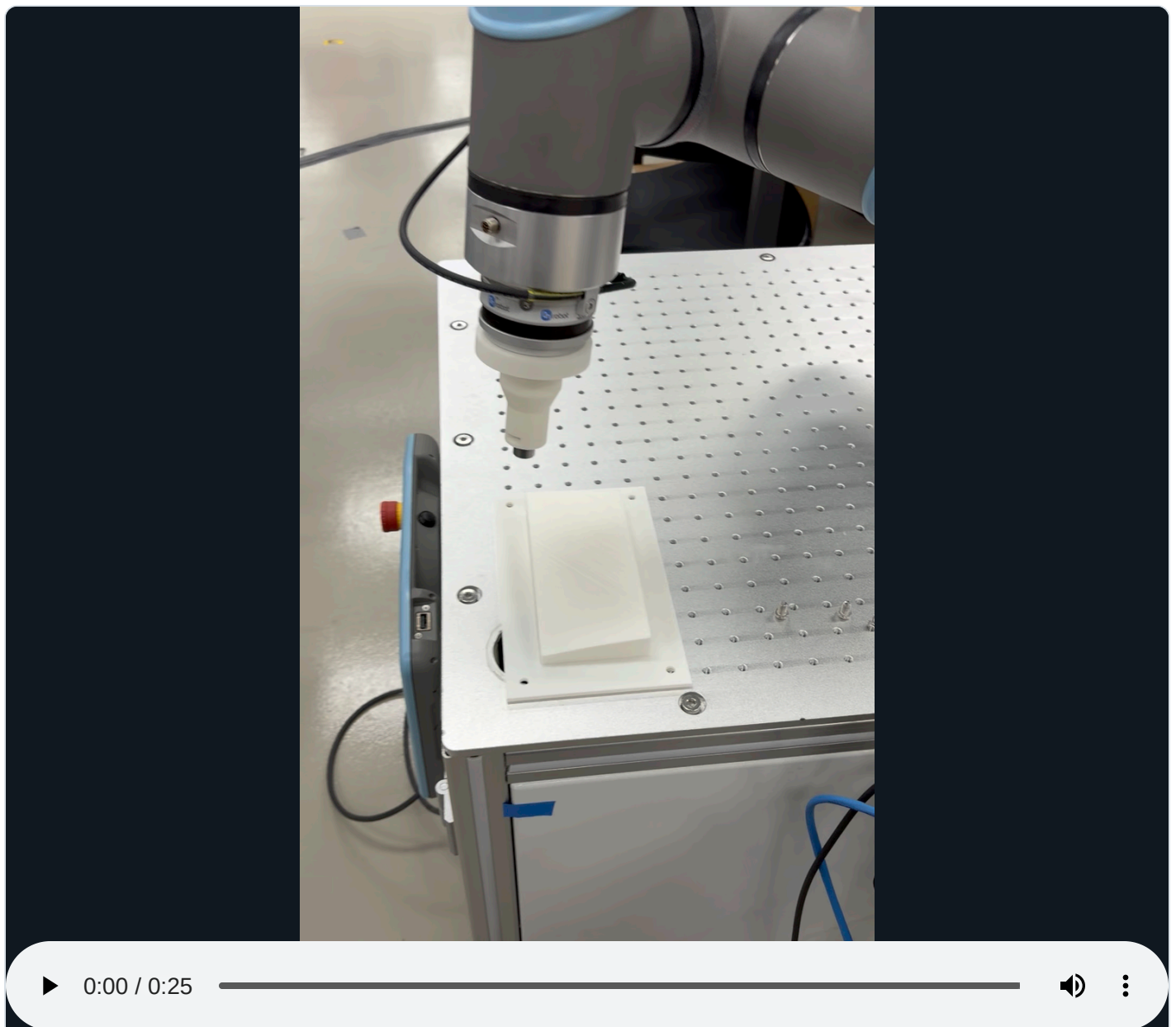


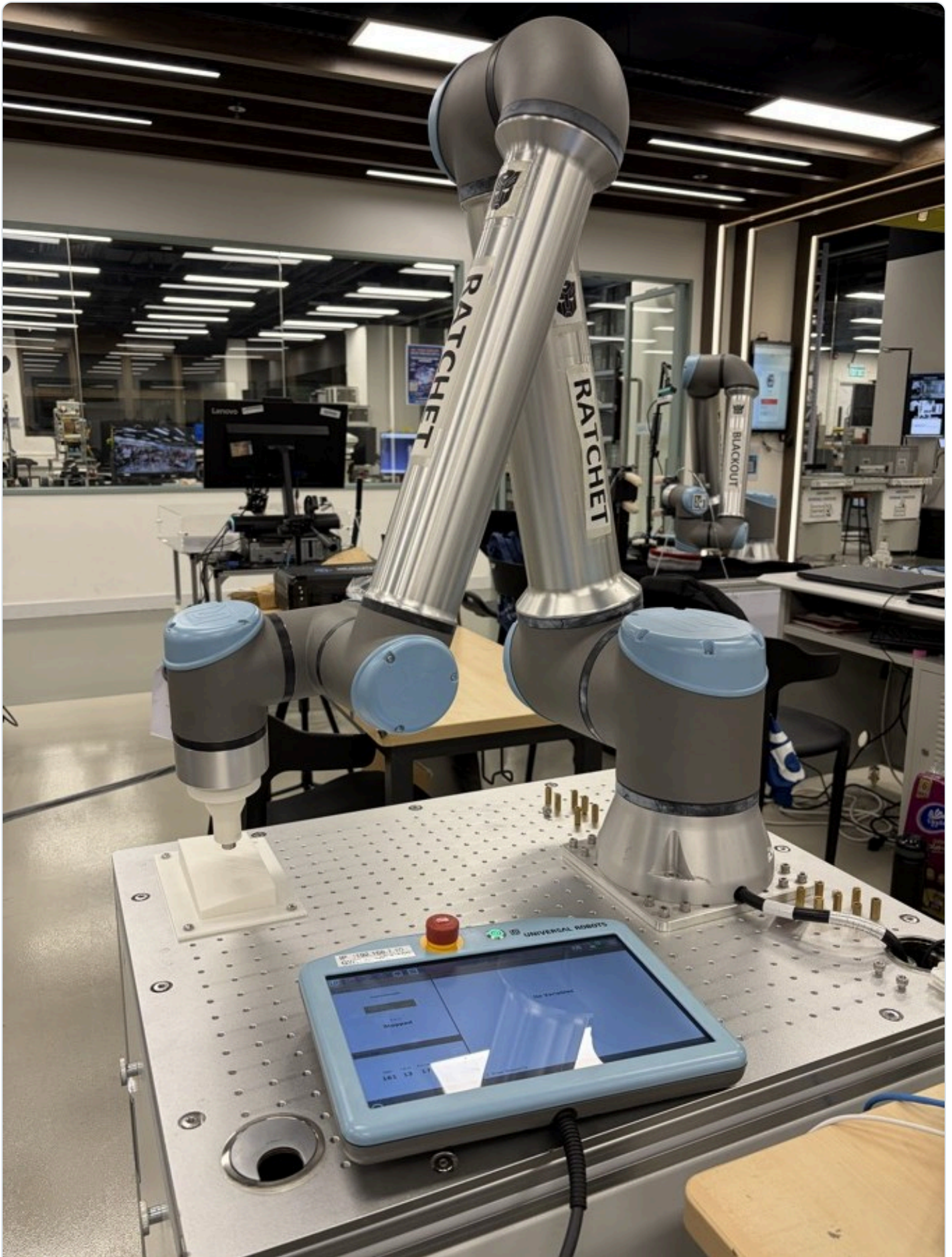
## BENCH SETUP

# UR10e / OnRobot HEX force-control bench

The meeting focus is the working experimental environment: the sensor is mounted, the final wiring state is visible, and the Compute Box / Ethernet / RTDE path is ready for repeatable force-control measurements.

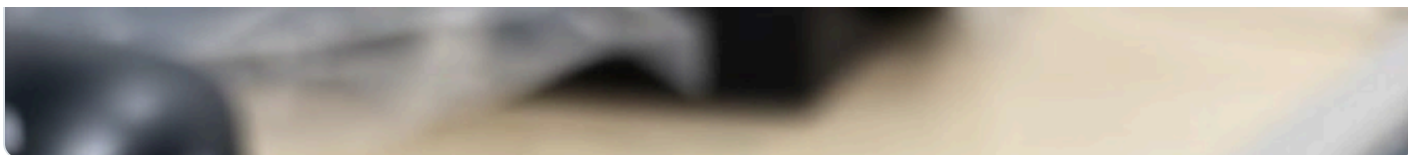


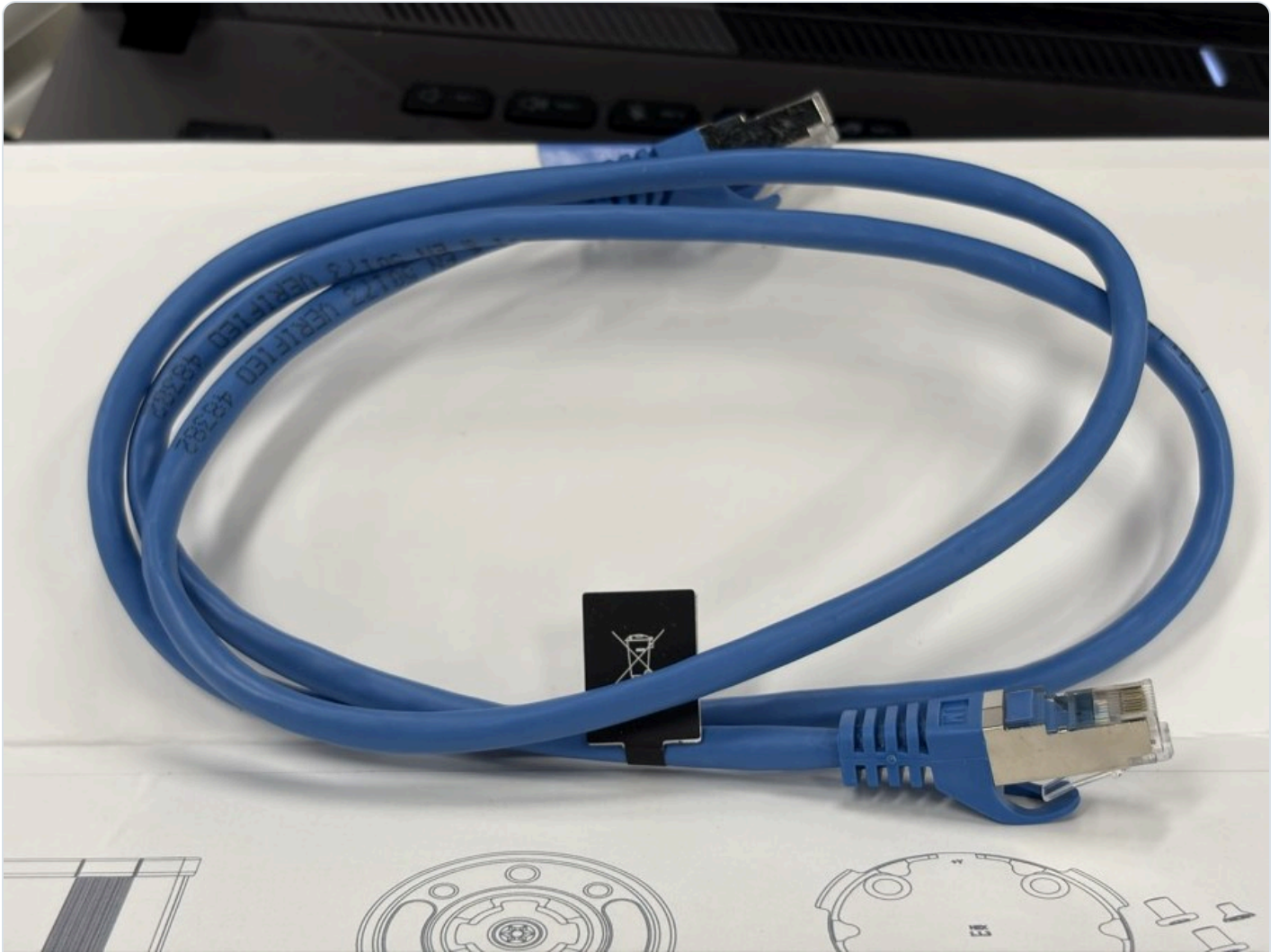
Final connected wiring state before the demo run.



UR10e bench context photo from setup.







Ethernet cable used in the Compute Box / switch path.

## DATA PATH

# Force data is now visible from the robot program into analysis plots

The bench uses the OnRobot URCap path for force-control commands and exports the measured wrench through RTDE output registers for Ubuntu-side logging.

## HEX sensor

Tool-side force/torque measurement



## Compute Box

Device cable, power, fixed IP, Ethernet



## Switch + UR10e + Ubuntu

Shared network for URCap discovery and RTDE logging



## RTDE registers

Fx/Fy/Fz/Tx/Ty/Tz available in the run log



## Meeting plots

Force tracking and path tracking from the same run window

The final sensor comparison should be remeasured under one zero-force protocol. Existing plots below are placeholders for layout and story flow.

### SENSOR BASELINE

## Baseline slot for force-sensor validation

First draft uses existing data-path plots as placeholders. The final version should actively zero force at the no-contact baseline pose, wait for the reading to settle, then

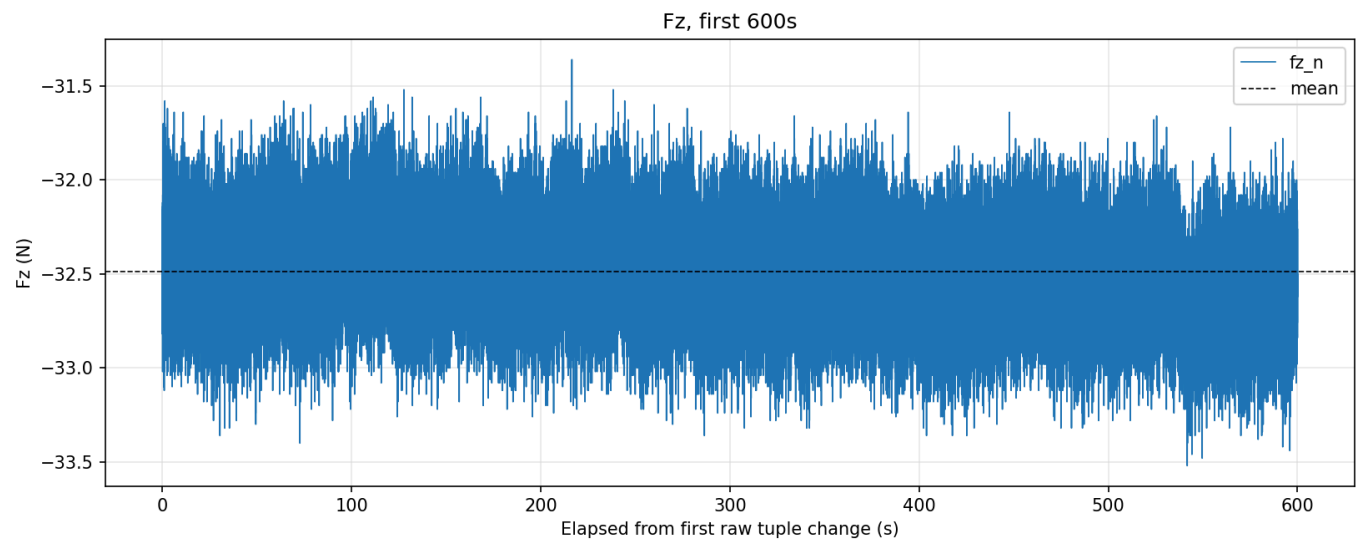
record OnRobot and UR built-in F/T under the same static condition.

### Baseline step

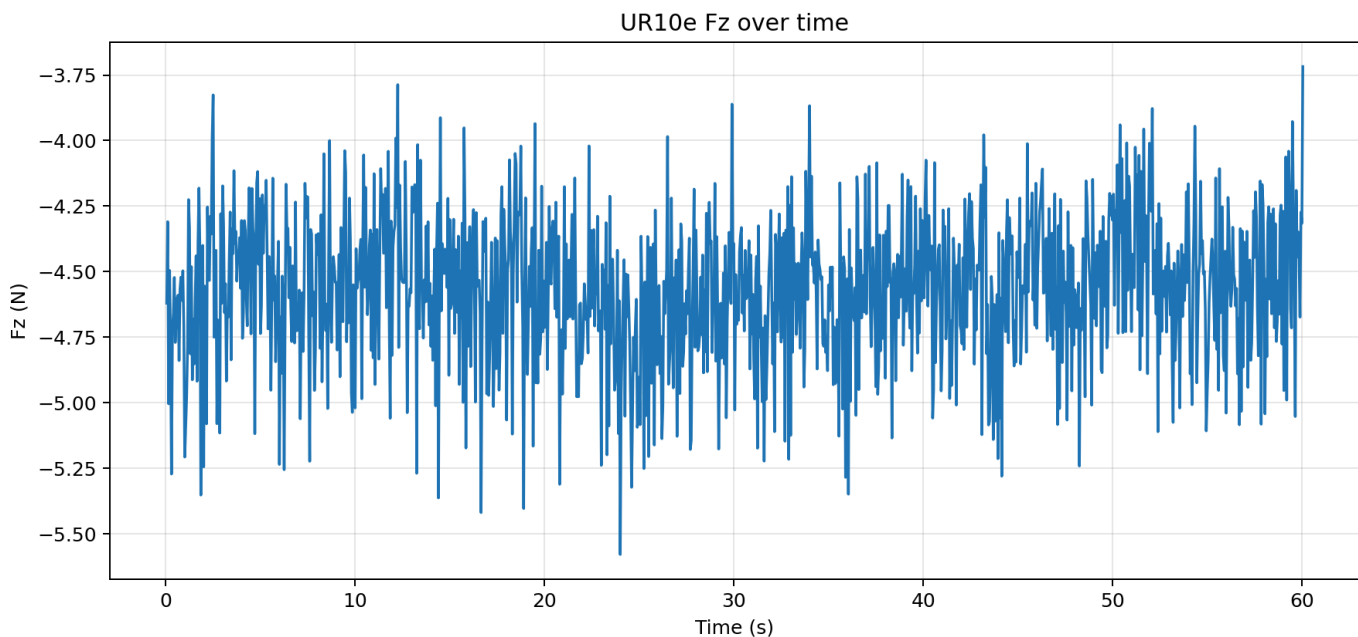
Move to the baseline slot, execute F/T Zero or the equivalent force-zero command, and only then start the static log.

### Comparison rule

Same zeroing step, same pose, same duration, same axis definitions, same time window.



OnRobot TCP DAQ Fz placeholder, 600 s. Not the final same-zero comparison.



UR built-in Fz placeholder, 60 s. Same-zero comparison pending.

## FORCE-CONTROL DEMO

# Closed-loop force tracking and contact path tracking

The demo result is evidence that the environment can run a force-controlled path. It is not the main story by itself.

**-5.000**

target Fz (N)

**0.612**

Fz error MAE (N)

**58.232**

Fz error IAE (N\*s)

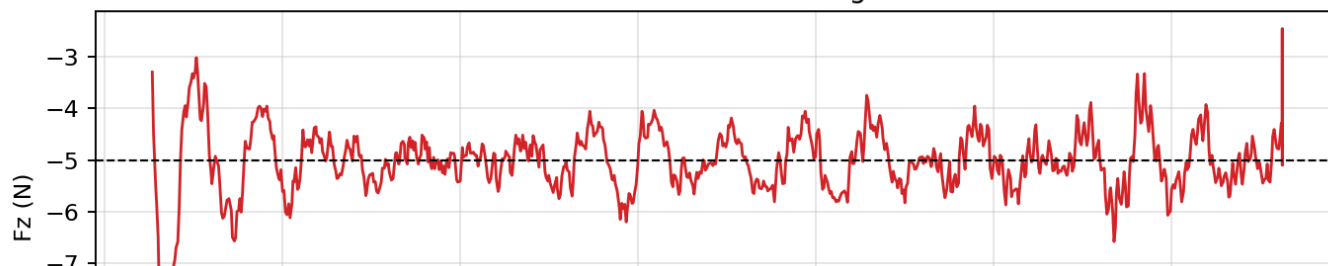
**85.9%**

within +/-1 N

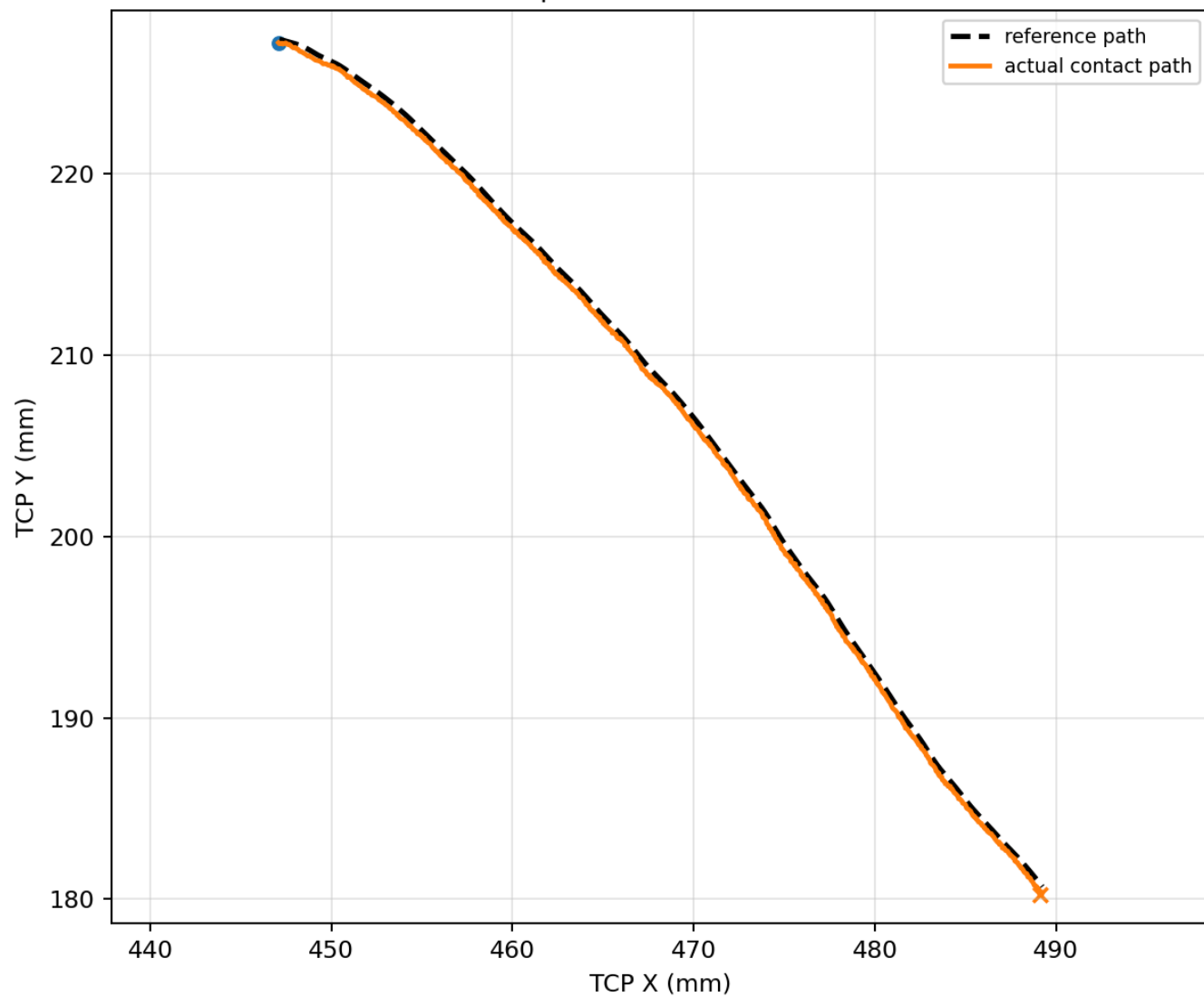
**0.304**

2D path P95 (mm)

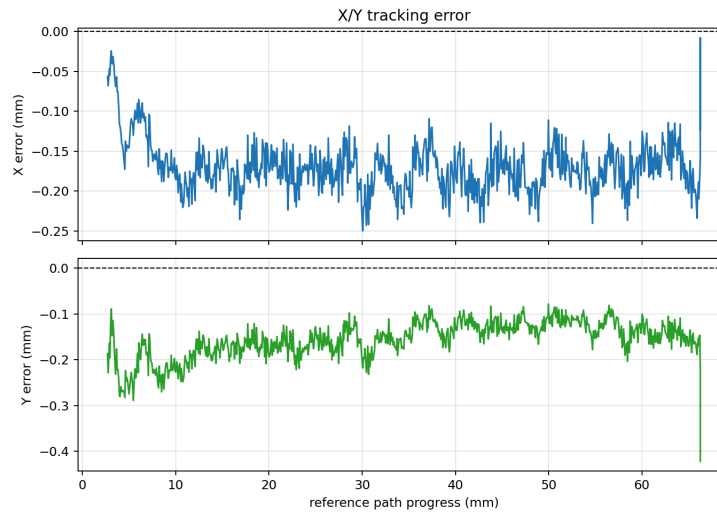
Measured Fz tracking



Contact path: actual vs reference



Actual contact path against reference path.



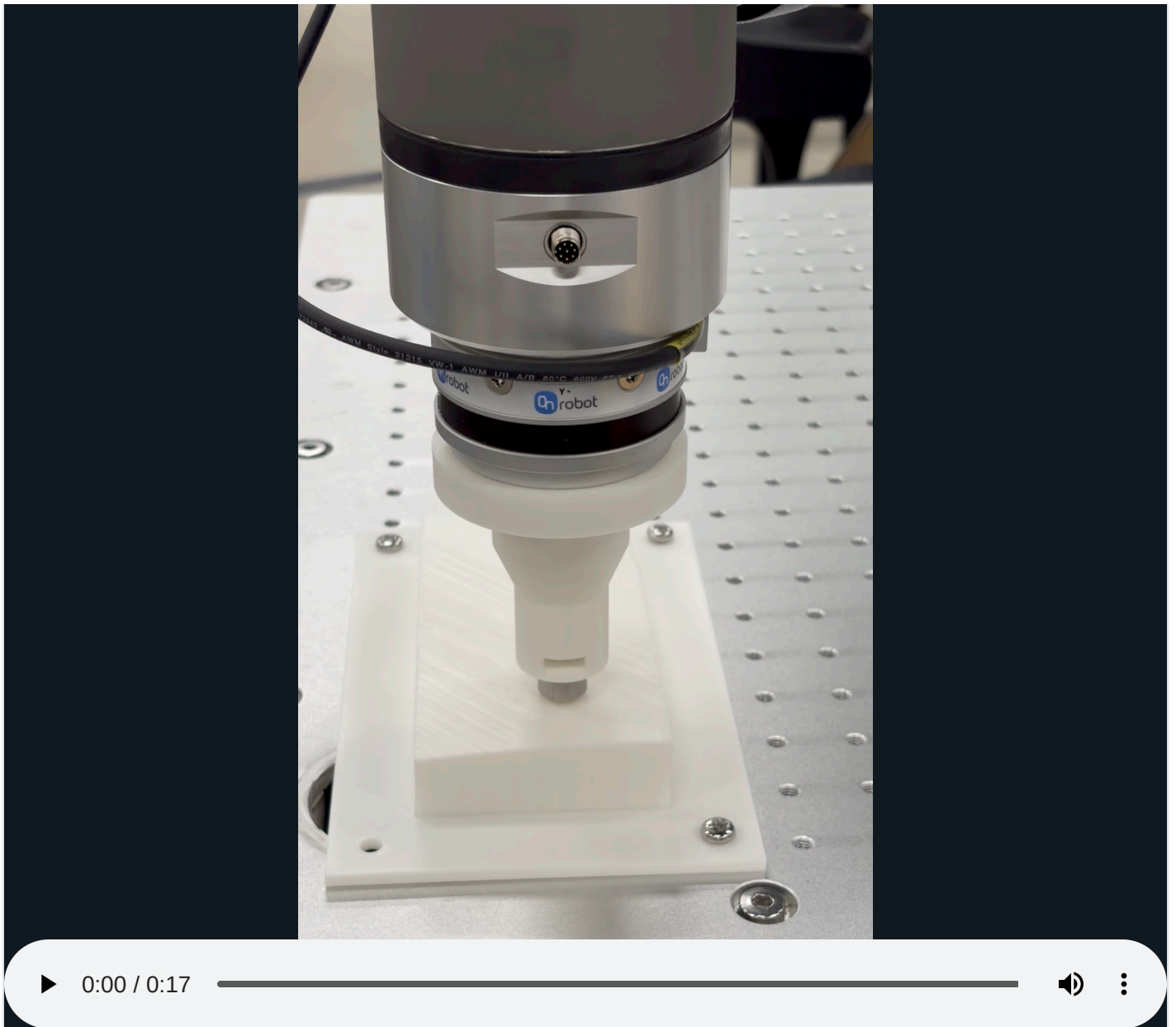
X/Y tracking error along the reference path.

## EXPERIMENT VIDEO

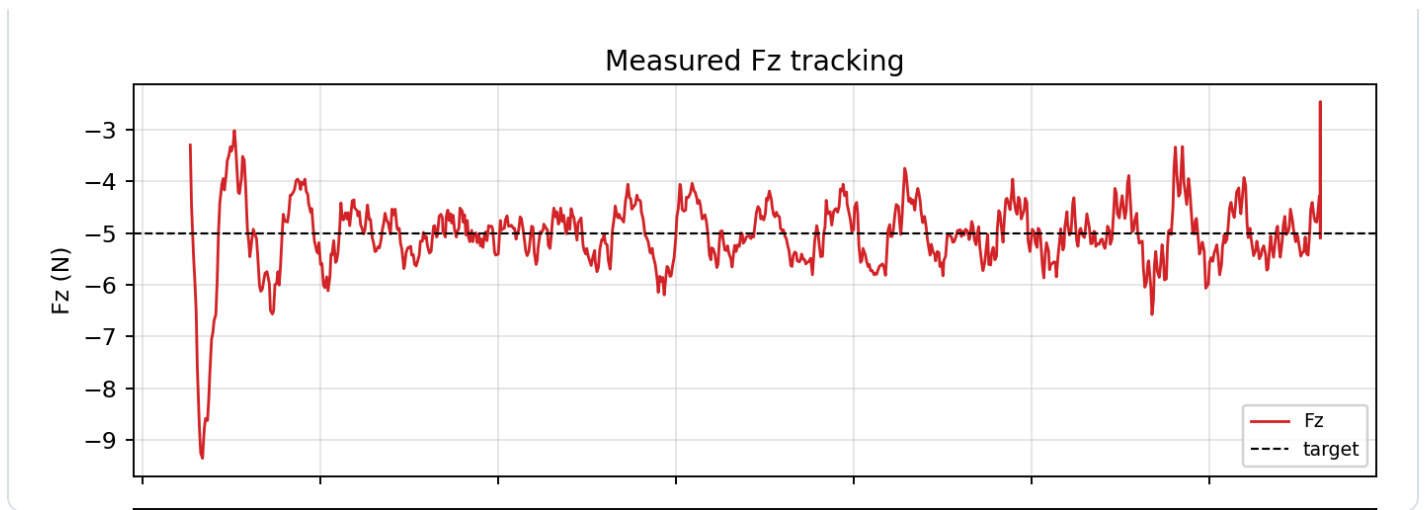
# Real robot contact demo video

This is the 29.2 MB robot experiment video. It is separate from the short PolyScope program-interface companion clip.





Robot contact experiment video.

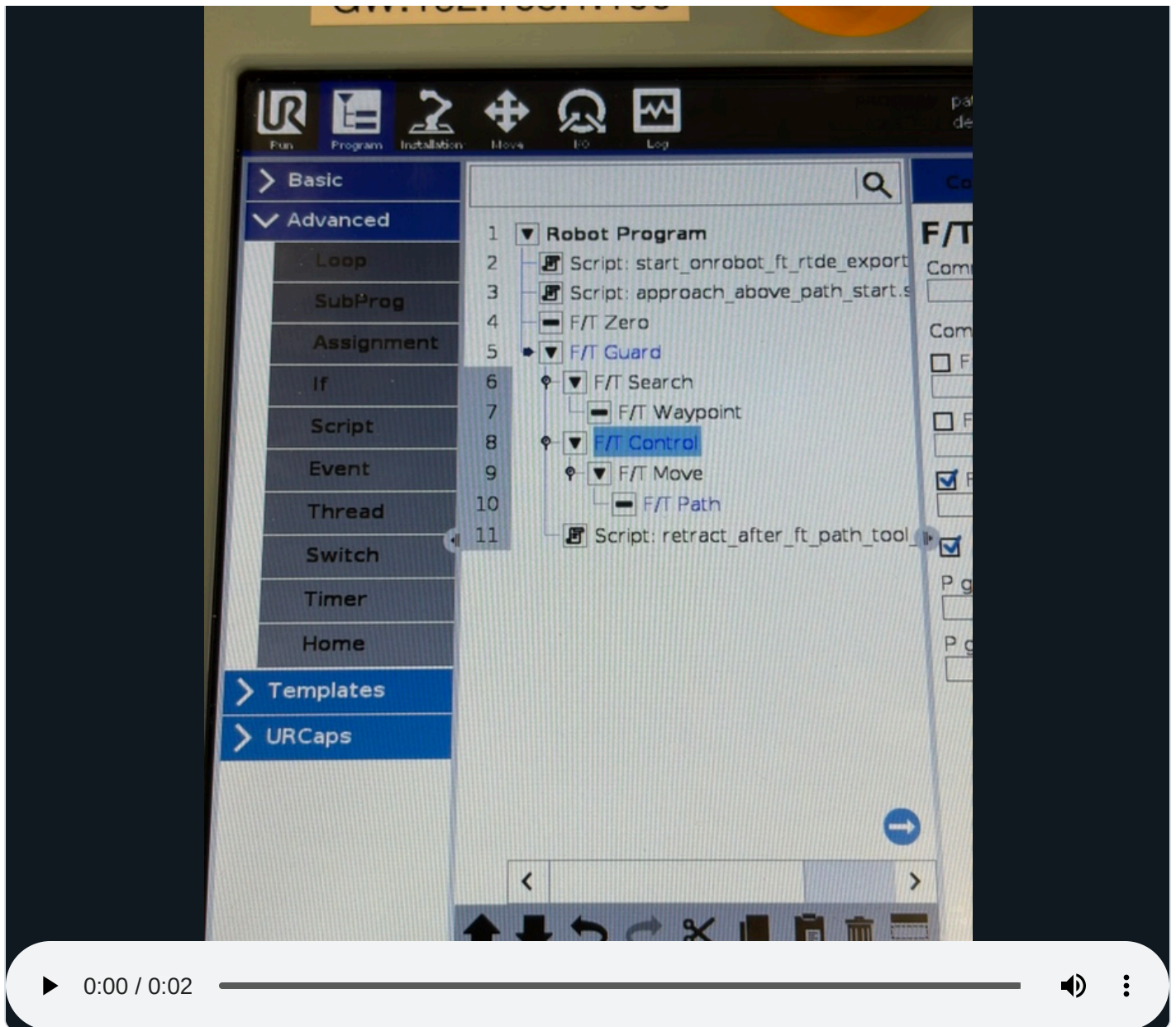


## PROGRAM INTERFACE

# URCap program structure

This clip is the PolyScope program interface, so it is named separately from the real robot demo video.





### Program nodes

F/T Zero, F/T Search, F/T Control, F/T Move, F/T Path.

### Settings kept in the meeting notes

Tool Z compliant, force PID, and F/T Move speed.

### Video separation

The robot experiment video is shown separately; this clip is only the program interface.